

Approximation $\rightarrow$ Approximation is a value or a number that is close to, but not exactly same.

$\{ +0.01 \text{ and } -0.01 \}$

$$\cdot 23.99 \approx 24$$

$$\cdot 45.99 \approx 46$$

$$\cdot 10.99 \approx 11$$

$$\cdot 56.99 \approx 57$$

$$\cdot 88.99 \approx 89$$

$$\cdot 73.99 \approx 74$$

$$\cdot 19.01 \approx 19$$

$$\cdot 72.01 \approx 72$$

$$\cdot 64.01 \approx 64$$

$$\cdot 31.01 \approx 31$$

$$\cdot 90.01 \approx 90$$

$$\cdot 85.01 \approx 85$$

A number can be approximated by rounding off.

Other Examples $\rightarrow$

$$\cdot 90.02 \approx 90$$

$$\cdot 53.04 \approx 53$$

$$\cdot 68.03 \approx 68$$

$$\cdot 27.05 \approx 27$$

$$\cdot 44.98 \approx 45$$

$$\cdot 71.97 \approx 72$$

$$\cdot 19.96 \approx 20$$

$$\cdot 85.95 \approx 86$$

$$\cdot 32.02 \approx 32$$

$$\cdot 59.93 \approx 60$$

Large Numbers $\rightarrow$

$$\cdot 4997 \approx 5000$$

$$\cdot 7401 \approx 7400$$

$$\cdot 9598 \approx 9600$$

$$\cdot 63998 \approx 64000$$

$$\cdot 123486 \approx 123500$$

$$\cdot 56998 \approx 57000$$

$$\cdot 24389 \approx 24400$$

$$\cdot 5397 \approx 5400$$

$$\cdot 128589 \approx 128600$$

Large Numbers

Approximation is usually used in case of Division or Percentage Calculation.

While solving Questions, choose the answer with nearest approximation.

Under Root Approximation $\rightarrow$

$$\bullet \sqrt{201} \longrightarrow 14.21$$

$$\begin{array}{cc} \swarrow & \searrow \\ \sqrt{196} & \sqrt{225} \end{array}$$

These are two nearest perfect squares.
Choose the most nearest i.e., $\sqrt{196}$.

Exact root of 201 will be 14.21 approximated to 14.



So, we will approximate the value under root to the nearest perfect square.

Hence,

$$\sqrt{201} \approx \sqrt{196} = 14.$$

Other similar examples $\rightarrow$

$$\bullet \sqrt{250} \approx \sqrt{256} = 16$$

$\swarrow$
Nearest perfect square.

$$\bullet \sqrt{173} \approx \sqrt{169} = 13$$

$$\bullet \sqrt{525} \approx \sqrt{529} = 23$$

$$\bullet \sqrt{630} \approx \sqrt{625} = 25$$

$$\bullet \sqrt{790} \approx \sqrt{784} = 28$$

$$\bullet \sqrt{955} \approx \sqrt{961} = 31$$

$$\bullet \sqrt{570} \approx \sqrt{576} = 24$$

$$\bullet \sqrt{450} \approx \sqrt{441} = 21$$

$$\bullet \sqrt{898} \approx \sqrt{900} = 30$$

Similarly, with cube roots $\rightarrow$

$$\bullet \sqrt[3]{66} \approx \sqrt[3]{64} = 4$$

$\swarrow$
Approximated

to nearest perfect cube root.

$$\bullet \sqrt[3]{130} \approx \sqrt[3]{125} = 5$$

$$\bullet \sqrt[3]{210} \approx \sqrt[3]{216} = 6$$

$$\bullet \sqrt[3]{350} \approx \sqrt[3]{343} = 7$$

$$\bullet \sqrt[3]{500} \approx \sqrt[3]{512} = 8$$

• Sum (2 Numbers) $\rightarrow$

Approximated to 15

$$24.07 + 15.03 \approx 24 + 15 = 39$$

Approximated
to 24

$$68.02 + 31.09 \approx 68 + 31 = 99$$

$$49.95 + 20.06 \approx 50 + 20 = 70$$

$$82.01 + 17.98 \approx 82 + 18 = 100$$

$$36.08 + 63.96 \approx 36 + 64 = 100$$

• Subtract (2 Numbers) $\rightarrow$

$$75.04 - 29.01 \approx 75 - 29 = 46$$

$$90.06 - 44.97 \approx 90 - 45 = 45$$

$$58.02 - 17.99 \approx 58 - 18 = 40$$

$$63.95 - 28.08 \approx 64 - 28 = 36$$

$$41.09 - 12.02 \approx 41 - 12 = 29$$

• Sum (3 Numbers) $\rightarrow$

$$12.07 + 25.09 + 41.02 \approx 12 + 25 + 41 = 78$$

$$33.98 + 16.06 + 50.01 \approx 34 + 16 + 50 = 100$$

$$24.03 + 39.95 + 18.08 \approx 24 + 40 + 18 = 82$$

$$71.04 + 14.97 + 9.06 \approx 71 + 15 + 9 = 95$$

$$60.02 + 27.08 + 12.99 \approx 60 + 27 + 13 = 100$$

• Subtract (3 Numbers) $\rightarrow$

$$100.07 - 39.98 - 20.02 \approx 100 - 40 - 20 = 40$$

$$88.01 - 17.96 - 10.09 \approx 88 - 18 - 10 = 60$$

$$64.02 - 14.99 - 20.06 \approx 64 - 15 - 20 = 29$$

Division →

$$\bullet \quad 31 \div 5 \Rightarrow \frac{31}{5} = 6.2 \longrightarrow \text{Approximated to } 6$$

$$\frac{31}{5} \approx \frac{30}{5} = 6 \checkmark$$

Direct Approximation in Question.

$$\bullet \quad 39.98 \div 5.01 = \frac{39.98}{5.01} \approx \frac{40}{5} = 8$$

$$\bullet \quad 71.96 \div 8.02 = \frac{71.96}{8.02} \approx \frac{72}{8} = 9$$

$$\bullet \quad 63.09 \div 6.01 = \frac{63.09}{6.01} \approx \frac{63}{6} = 10.5$$

$$\bullet \quad 89.97 \div 9.03 = \frac{89.97}{9.03} \approx \frac{90}{9} = 10$$

Multiplication →

In Multiplication, You should be careful in the cases where both the numbers are above their base values (both greater than 0.00) and where both are below their base values (both less than 0.00).

like,

This should not be approximated

$$190.21 \times 150.18$$

$$\neq 190 \times 150 \quad \text{like this}$$

Because,

{ both are greater than
Base values 190 & 150 }

$$190.21 \times 150.18 = \underline{28565}$$

$$190 \times 150 = \underline{28500}$$

This can create a large error.

There is a difference of 65.

However, the safer approximation will be when one number is slightly above 0.00 and another is below 0.00

"One up and One Down" case.

for example,

$\underbrace{90.11} \times \underbrace{49.95} \rightarrow$ This is slightly below the approximated value 50.
 $\downarrow$
 This is above the approximated value 90

$$\begin{aligned} \text{Hence, } 90.11 \times 49.95 &= \underline{4500.9945} \\ 90 \times 50 &= \underline{4500} \end{aligned}$$

Not creating much difference in answer

More Similar Examples $\rightarrow$

$$\bullet \quad \underbrace{18.07} \times \underbrace{12.96} \approx \underbrace{18} \times \underbrace{13} = 234$$

$$\bullet \quad 32.02 \times 25.03 \longrightarrow 32.02 \times 25.03 = 801.4$$

$$32 \times 25 = 800$$

Both numbers are above approximated value 32 & 25

But the answer does not create a larger difference

$$\bullet \quad 45.09 \times 19.92 \approx 45 \times 20 = 900$$

$$\bullet \quad 27.04 \times 15.02 \approx 27 \times 15 = 405$$

What Approximate value should come in the place of question mark (?) in the following question?

निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर अनुमानित क्या आना चाहिए?

Q1. $(? + 17.98) = 24.98\%$ of 80.01

- (A) 0 (B) 2 (C) 4 (D) 6 (E) 8

$$x + 17.98 = 24.98\% \text{ of } 80.01$$

$$x + 18 = 25\% \text{ of } 80 \quad \{ \text{Approximation} \}$$

$$x + 18 = \frac{1}{4} \times 80 \Rightarrow x + 18 = 20$$

$$x = 2$$

Q2. $(35.89 \div 5.99) + ? = 19.97$

- (A) 10 (B) 12 (C) 14 (D) 16 (E) 18

$$(35.89 \div 5.99) + x = 19.97$$

$$(36 \div 6) + x = 20$$

$$\frac{36}{6} + x = 20$$

$$6 + x = 20 \Rightarrow x = 20 - 6 \Rightarrow x = 14$$

Q3. $(? \div 7.98) = 6.69\%$ of 75

- (A) 40 (B) 45 (C) 50 (D) 55 (E) 60

$$(x \div 7.98) = 6.69\% \text{ of } 75$$

$$x \div 8 = 6.66\% \text{ of } 75$$

$$\frac{x}{8} = \left(\frac{1}{15} \times 75 \right)$$

$$x = 5 \times 8 \Rightarrow x = 40$$

$$\begin{array}{ccc} & 6.69\% & \\ \swarrow & & \searrow \\ \frac{1}{16} & 6.25\% & 6.66\% \frac{1}{15} \end{array}$$

Two options

→ Nearest option will be 6.66%

Q4. $(63.79 \div 7.89) + 19.90\%$ of ? = 20.10

- (A) 40 (B) 45 (C) 50 (D) 55 (E) 60

$$(63.79 \div 7.89) + (19.90\% \text{ of } x) = 20.10$$

$$(64 \div 8) + (20\% \text{ of } x) = 20$$

$$\left(\frac{64}{8} \right) + \left(\frac{1}{5} \times x \right) = 20$$

$$\frac{x}{5} = 20 - 8 \Rightarrow x = 12 \times 5 \Rightarrow x = 60$$

Q5. $\left(\frac{3.01}{3.99} \text{ of } 95.99\right) - ? = 35.98$

(A) 30

(B) 32

(C) 34

☒ (D) 36

(E) 38

$$\left(\frac{3.01}{3.99} \text{ of } 95.99\right) - x = 35.98$$

$$\left(\frac{3}{4} \times 96\right) - x = 36$$

$$72 - x = 36 \Rightarrow x = 72 - 36$$

$$x = 36$$

Q6. $(12.48\% \text{ of } ?) + 7.99 = 40.04$

(A) 224

☒ (B) 256

(C) 284

(D) 302

(E) 312

$$(12.48\% \text{ of } x) + 7.99 = 40.04$$

$$(12.5\% \text{ of } x) + 8 = 40$$

$$\left(\frac{1}{8} \times x\right) = 40 - 8$$

$$\frac{x}{8} = 32 \Rightarrow x = 32 \times 8$$

$$x = 256$$

Q7. $(? \div 6.03) + (18.01 \div 2.99) = 13.99$

(A) 36

(B) 42

☒ (C) 48

(D) 54

(E) 60

$$\left(\frac{x}{6.03}\right) + \left(\frac{18.01}{2.99}\right) = 13.99$$

$$\frac{x}{6} + \frac{18}{3} = 14$$

$$\frac{x}{6} = 14 - 6 \Rightarrow x = 8 \times 6$$

$$x = 48$$

Q8. $(44.98\% \text{ of } 199) - ? = 49.99$

(A) 35

☒ (B) 40

(C) 45

(D) 50

(E) 60

$$(44.98\% \text{ of } 199) - x = 49.99$$

$$(45\% \text{ of } 200) - x = 50$$

$$\left(\frac{45}{100} \times 200\right) - x = 50$$

$$90 - x = 50 \Rightarrow x = 90 - 50$$

$$x = 40$$

Q9. $(83.98 \div 6.97) + (? \times 9.09\%) = 20$

(A) 71

(B) 76

(C) 82

☒ (D) 88

(E) 96

$$(83.98 \div 6.97) + (x \times 9.09\%) = 20$$

$$\left(\frac{84}{7}\right) + \left(x \times \frac{1}{11}\right) = 20$$

$$12 + \frac{x}{11} = 20 \implies \frac{x}{11} = 20 - 12$$

$$x = 8 \times 11 \implies \underline{x = 88} \checkmark$$

Q10. $\{(? - 29.98) \times 11.13\} + 5.99 = 11.96$

(A) 68

(B) 72

(C) 76

(D) 80

☒ (E) 84

$$\{(x - 30) \times 11.11\} + 6 = 12$$

$$(x - 30) \times \frac{1}{9} = 6$$

$$(x - 30) = 54$$

$$x = 54 + 30$$

$$x = 84$$

SIMPLIFICATION MISCELLANEOUS QUESTIONS

What value will come in place of question mark (?) in the following question?

Q1. $3.75\% \text{ of } 160 + 3.75\% \text{ of } 240 = ?\% \text{ of } 20$

(A) 60

(B) 70

(C) 80

☒ (D) 75

(E) None of these

$$(3.75\% \text{ of } 160) + (3.75\% \text{ of } 240) = x\% \text{ of } 20$$

$$3.75\% \text{ of } [160 + 240] = x\% \text{ of } 20$$

$$\frac{15}{400} \times 400 = \frac{x}{100} \times 20$$

$$15 = \frac{x}{5} \implies x = 15 \times 5$$

$$x = 75$$

$$\left. \begin{aligned} 3.75\% \\ 3.75 &= 3 + 0.75 \\ &= 3 + \frac{3}{4} \\ &= \frac{15}{4} \\ 3.75\% &= \frac{15}{400} \end{aligned} \right\}$$

Q2. $\frac{7}{18} - \left[11 - \left(\frac{35}{54} - \frac{28}{9}\right)\right] = ?$

☒ (A) -353/27

(B) 353/27

(C) -353/21

(D) -343/27

(E) None of these

$$\frac{7}{18} - \left[11 - \left(\frac{35 \times 1 - 28 \times 6}{54}\right)\right]$$

{ Taking LCM }

$$\Rightarrow \frac{7}{18} - \left[11 - \left(-\frac{133}{54} \right) \right] \Rightarrow \frac{7}{18} - \left[11 + \frac{133}{54} \right]$$

$$\Rightarrow \frac{7}{18} - \left[\frac{11 \times 54 + 133 \times 1}{54} \right] \Rightarrow \frac{7}{18} - \left[\frac{594 + 133}{54} \right]$$

$$\Rightarrow \frac{7}{18} - \frac{727}{54} \Rightarrow \frac{21 - 727}{54} \Rightarrow -\frac{706}{54} \Rightarrow -\frac{353}{27} \checkmark$$

Q3.

$$\frac{0.042 \times 0.06}{0.00084} + \frac{0.315 \times 0.8}{0.0035} = ?$$

(A) 85

(B) 70

(C) 80

(D) 65

(E) None of these

Ans $\rightarrow 75$

$$\Rightarrow \left[\frac{\frac{42}{1000} \times \frac{6}{100}}{84} \right] + \left[\frac{\frac{315}{1000} \times \frac{8}{10}}{35} \right]$$

$$\Rightarrow \left(\frac{42 \times 6}{84} \right) + \left(\frac{315 \times 8}{35} \right)$$

$$\Rightarrow 3 + 72 \Rightarrow 75$$

$$Q4. \left\{ 1 \div \left(\frac{1}{5} + \frac{1}{10} \right) \right\} + \left\{ 1 \div \left(\frac{1}{3} + \frac{1}{15} \right) \right\} = ?$$

(A) 25/6

(B) 5/6

(C) 35/5

(D) 30/6

Ans $\rightarrow \frac{35}{6}$
(E) None of these

$$\Rightarrow \left\{ 1 \div \left(\frac{2+1}{10} \right) \right\} + \left\{ 1 \div \left(\frac{5+1}{15} \right) \right\}$$

{Taking LCM}

Solving Brackets

Inwards to outwards

$$\Rightarrow \left\{ 1 \div \frac{3}{10} \right\} + \left\{ 1 \div \frac{6}{15} \right\}$$

$$\Rightarrow \left\{ 1 \times \frac{10}{3} \right\} + \left\{ 1 \times \frac{15}{6} \right\} \quad \left\{ \text{Reciprocal and sign change} \right\}$$

$$\Rightarrow \frac{10}{3} + \frac{5}{2} \Rightarrow \frac{20+15}{6} \Rightarrow \frac{35}{6} \checkmark$$

{Taking LCM, LCM of 2 & 3 will be 6}

As like, a% of b = b% of a

→ 432% of 25 can be
25% of 432

Q5. $(? - 18)\%$ of 900 = 432% of 25

(A) 48

(B) 36

(C) 42

(D) 24

(E) None of these

Ans → 30

$$\frac{x-18}{100} \times 900 = 25\% \text{ of } 432$$

$$(x-18)9 = \frac{1}{4} \times 432$$

$$x-18 = \frac{108}{9} \Rightarrow x-18 = 12$$

$$x = 12 + 18$$

$$x = 30$$

Q6.

$6^{-2} + 97\frac{2}{9}\%$ of 361 = ?

(A) 360

(B) 350

(C) 351

(D) 352

(E) None of these

$$\Rightarrow \left(\frac{1}{6^2}\right) + \left[\left(\frac{875}{9} \times \frac{1}{100}\right) \times 361\right]$$

Mixed fraction

$$97\frac{2}{9} = \frac{(97 \times 9) + 2}{9}$$

$$= \frac{873+2}{9} = \frac{875}{9}$$

$$\Rightarrow \left(\frac{1}{36}\right) + \left(\frac{35}{36} \times 361\right)$$

$$\Rightarrow \frac{1}{36} \left[1 + (35 \times 361)\right] \Rightarrow \left(\frac{1}{36} + \frac{35 \times 361}{36}\right) \text{ \{Taking LCM\}}$$

$$\Rightarrow \frac{1 + (35)(361)}{36} \Rightarrow \frac{1 + (35)(360+1)}{36} \Rightarrow \frac{1 + (35)(360) + (35)(1)}{36}$$

$$\Rightarrow \frac{36 + (35)(360)}{36} \Rightarrow \frac{36}{36} + \frac{(35)(360)}{36} \Rightarrow 1 + (35)(10)$$

$$\Rightarrow 1 + 350$$

$$\Rightarrow 351$$

Q7.

$789.65 + 210.35 + 99.90 + 0.10 + 500 = ?$

(A) 1620

(B) 1600

(C) 1550

(D) 1580

(E) None of these

$$\Rightarrow (789.65 + 210.35) + (99.90 + 0.10) + (500)$$

$$\Rightarrow 1000 + 100 + 500$$

$$\Rightarrow 1000 + 600$$

$$\Rightarrow 1600$$

{ Simple Addition
in Decimal }

Q8.

$$\left(\frac{1}{3125}\right)^{-\frac{3}{5}} + \left(\frac{1}{4096}\right)^{-\frac{3}{4}} = ? \times 7$$

(A) 98

(B) 91

(C) 105

(D) 84

(E) None of these

$$\left(\frac{1}{3125}\right)^{-\frac{3}{5}} + \left(\frac{1}{4096}\right)^{-\frac{3}{4}} = x \times 7$$

$$(3125)^{\frac{3}{5}} + (4096)^{\frac{3}{4}} = x \times 7$$

$$(5^5)^{\frac{3}{5}} + (8^4)^{\frac{3}{4}} = x \times 7$$

$$5^3 + 8^3 = x \times 7$$

$$125 + 512 = x \times 7 \Rightarrow 7x = 637$$

$$x = \frac{637}{7} \Rightarrow \underline{91} \checkmark$$

Q9.

$$\sqrt{48^4} + 1152 = ? \times 24^2$$

(A) 5

(B) 6

(C) 8

(D) 4

(E) None of these

$$(48^4)^{\frac{1}{2}} + 1152 = x \times 24^2$$

$$(48)^{4 \times \frac{1}{2}} + 1152 = x \times 576$$

$$48^2 + 1152 = 576x$$

$$2304 + 1152 = 576x$$

$$576x = 3456 \Rightarrow x = \frac{3456}{576} \Rightarrow \underline{x=6} \checkmark$$

Q10. 18.75% of $1696 + (6.4 \times 3.75 \times 20) = ? + 258$

(A) 552

(B) 560

(C) 528

(D) 540

(E) None of these

$$\left(\frac{3}{16} \times 1696\right) + (6.4 \times 20 \times 3.75) = x + 258$$

$$(3 \times 106) + (128 \times \frac{15}{4}) = x + 258$$

$$318 + 480 - 258 = x$$

$$540 = x$$

$$\checkmark 18.75\% = 6.25\% \times 3$$

$$= \frac{1}{16} \times 3$$

$$= \frac{3}{16}$$

$$\checkmark 3.75 = 3 + \frac{3}{4} = \frac{15}{4}$$

Q11. $64^2 \times 16^x \div 8^5 = 2^{19}$

(A) 6

(B) 5

(C) 4.5

(D) 6.5

(E) None of these

$$64^2 \times 16^x \div 8^5 = 2^{19}$$

$$(2^6)^2 \times (2^4)^x \div (2^3)^5 = 2^{19}$$

$$2^{12} \times 2^{4x} \div 2^{15} = 2^{19}$$

$$2^{(12+4x-15)} = 2^{19} \Rightarrow 12+4x-15=19$$

$$4x = 19+3$$

$$x = \frac{22}{4} = \frac{11}{2} = 5.50$$

Q12. $58^2 + 5^3 - \sqrt{1089} = 2 \times x^3$

(A) 14

(B) 12

(C) 10

(D) 8

(E) None of these

$$58^2 + 5^3 - \sqrt{1089} = 2 \times x^3$$

$$3364 + 125 - 33 = 2 \times x^3$$

$$3456 = 2 \times x^3$$

$$1728 = x^3$$

$$x = \sqrt[3]{1728} \Rightarrow \underline{x = 12}$$

$$\left\{ \begin{array}{l} 58^2 = 3364 \\ 5^3 = 125 \\ \sqrt{1089} = 33 \end{array} \right\}$$

Q13. $\left(\frac{1}{3} + \frac{1}{4}\right) \div \left(\frac{5}{6} - \frac{1}{2}\right) = ?$

(A) 7/4

(B) 3/2

(C) 5/4

(D) 7/3

(E) None of these

$$\Rightarrow \left(\frac{1}{3} + \frac{1}{4}\right) \div \left(\frac{5}{6} - \frac{1}{2}\right) \quad \{\text{Taking LCM}\}$$

$$\Rightarrow \left(\frac{4+3}{12}\right) \div \left(\frac{5-3}{6}\right)$$

$$\Rightarrow \frac{7}{12} \div \frac{2}{6} \Rightarrow \frac{7}{12} \times \frac{6}{2} \Rightarrow \frac{7}{4}$$

Q14. $5^4 \div 5^6 \times 125 \times 14.4 + 80\% \text{ of } 25 = ?$

(A) 92

(B) 88

(C) 90

(D) 94

(E) None of these

$$\left[\left(\frac{5^4}{5^6} \times 5^3\right) \times \frac{144}{10}\right] + (25 \cdot 80\% \text{ of } 80)$$

$$\Rightarrow \left(5^{(4-6+3)} \times \frac{144}{10} \right) + \left(\frac{1}{4} \times 80 \right)$$

$$\Rightarrow \left(5 \times \frac{144}{10} \right) + (20) \Rightarrow \frac{144}{2} + 20$$

$$\Rightarrow 72 + 20 \Rightarrow 92$$

Q15. $(\sqrt{0.49} \div 0.07) + (0.18 \div 0.006) = ?$

(A) 38

(B) 40

(C) 42

(D) 44

(E) None of these

$$\Rightarrow \left[\left(\frac{49}{100} \right)^{\frac{1}{2}} \div \left(\frac{7}{100} \right) \right] + \left(\frac{18}{100} \div \frac{6}{1000} \right)$$

$$\Rightarrow \left[\left(\frac{7}{10} \right)^{2 \times \frac{1}{2}} \div \frac{7}{100} \right] + \left(\frac{18}{100} \div \frac{6}{1000} \right)$$

$$\Rightarrow \left(\frac{7}{10} \times \frac{100}{7} \right) + \left(\frac{18}{100} \times \frac{1000}{6} \right) \Rightarrow (10) + (30)$$

$$\Rightarrow 40$$

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निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर अनुमानित क्या आना चाहिए?

Q11: $11.97 + 12.02 - 3.03 \times 3.99 = ?$

(A) 12

(B) 15

(C) 10

(D) 9

(E) None of these

$$\Rightarrow 11.97 + 12.02 - (3.03 \times 3.99)$$

$$\Rightarrow 12 + 12 - (3 \times 4) \quad (\text{Approximation})$$

$$\Rightarrow 12 + 12 - 12$$

$$\Rightarrow 12$$

Q12: $118.1 + 212.01 + 99.99 = ? + 311.97$

(A) 118

(B) 120

(C) 124

(D) 126

(E) 130

$$\Rightarrow 118 + (212 + 100) = x + 312 \quad (\text{Approximation})$$

$$\Rightarrow 118 + 312 - 312 = x$$

$$\Rightarrow x = 118$$

Q13: $139.97 - 29.97 \times 60.03 \div 14.99 = ?$

(A) 440

☒ (B) 20

(C) 40

(D) 220

(E) 100

$$\Rightarrow (139.97) - (29.97 \times 60.03 \div 14.99)$$

$$\Rightarrow (140) - (30 \times 60 \div 15)$$

$$\Rightarrow (140) - \left(\frac{30 \times 60}{15} \right)$$

$$\Rightarrow 140 - 120$$

$$\Rightarrow 20$$

Q14: 13.97% of $111.98 \div 195.95 = ?\%$

(A) 7

☒ (B) 8

(C) 9

(D) 10

(E) 12

$$\Rightarrow 13.97\% \text{ of } 111.98 \div 195.95 = x\%$$

$$\Rightarrow 14\% \text{ of } 112 \div 196 = x\%$$

$$\Rightarrow \frac{14}{100} \times 112 \times \frac{1}{196} = \frac{x}{100}$$

$$\Rightarrow \frac{112}{14} = x \Rightarrow \underline{x = 8} \checkmark$$

Q15: $55.03 \div 0.01 \div 99.99 = ?$

(A) 0.55

(B) 5.50

☒ (C) 55

(D) 550

(E) 5500

$$\Rightarrow (55.03) \div (0.01) \div (99.99)$$

$$\Rightarrow (55) \div (0.01) \div (100)$$

$$\Rightarrow \frac{(55)}{\left(\frac{1}{100} \right)} \div (100)$$

$$\Rightarrow 55 \times \cancel{100} \times \frac{1}{\cancel{100}} \Rightarrow 55$$

Q16. $(48.9 \times 19.8 \div 9.7) \div 98.03 = ?$

(A) 1

(B) 2

(C) 3

(D) 4

(E) 5

$$\Rightarrow (48.9 \times 19.8 \div 9.7) \div (98.03)$$

$$\Rightarrow (49 \times 20 \div 10) \div (98)$$

$$\Rightarrow 49 \times 20 \times \frac{1}{10} \times \frac{1}{98} \Rightarrow \frac{49 \times 2}{98} \Rightarrow \frac{98}{98} \Rightarrow 1$$

{Reciprocal and sign "÷" to "x"}

Q17. $\{(62.03 \div 3.1) - 4.9\} \div 4.9 = ?$

(A) 1

(B) 2

(C) 3

(D) 4

(E) 5

$$\Rightarrow \{(62.03 \div 3.1) - 4.9\} \div 4.9$$

$$\Rightarrow \{(62 \div 3.1) - 5\} \div 5 \quad (\text{Approximation})$$

$$\Rightarrow \left(\frac{62}{3.1} - 5\right) \div 5$$

$$\Rightarrow (20 - 5) \div 5 \Rightarrow \frac{15}{5} \Rightarrow 3$$

$$\left\{ \begin{aligned} \frac{62}{3.1} &\approx \frac{62 \times 10}{31} \\ &\Rightarrow 2 \times 10 \\ &\Rightarrow 20. \end{aligned} \right\}$$

Q18. $\left(\frac{7}{9} \text{ of } 323.80\right) + 11.6 = (? \div 1.98)$

(A) 528

(B) 480

(C) 576

(D) 600

(E) 484

$$\Rightarrow \left(\frac{7}{9} \times 323.8\right) + 11.6 = x \div 1.98$$

$$\Rightarrow \left(\frac{7}{9} \times 324\right) + 12 = x \div 2$$

$$\Rightarrow (7 \times 36) + 12 = \frac{x}{2}$$

$$\Rightarrow (252 + 12) \times 2 = x$$

$$264 \times 2 = x$$

$$x = 264 \times 2$$

$$x = 528 \checkmark$$

Q19. $(? - 13.99) \div 2.51 = \left(\frac{1}{6} \text{ of } 151\right) - 18.99$

(A) 1

(B) 15

(C) 14

(D) 19

(E) 29

$$\Rightarrow (x - 13.99) \div 2.51 = \left(\frac{1}{6} \times 151\right) - 18.99$$

$$\Rightarrow \frac{x - 14}{2.5} = \left(\frac{1}{6} \times 150\right) - 19$$

$$\Rightarrow \frac{x - 14}{2.5} = 25 - 19$$

$$\Rightarrow x - 14 = 6 \times 2.5$$

$$\begin{aligned} \Rightarrow x - 14 &= 15.0 \\ x &= 15 + 14 \\ x &= 29 \end{aligned}$$

Q20. $(? \times 14.30\%) + (63.99 \div 3.99) = 31.99$

(A) 98

(B) 105

(C) 112

(D) 119

(E) 126

14.30% $\rightarrow$ Approximated to
14.28% = $\frac{1}{7}$

{ Explained in
Percentage class }

$$(x \times 14.30\%) + (63.99 \div 3.99) = 31.99$$

$$(x \times 14.28\%) + (64 \div 4) = 32$$

$$\left(x \times \frac{1}{7}\right) + \left(\frac{64}{4}\right) = 32$$

$$\frac{x}{7} + 16 = 32$$

$$\frac{x}{7} = 16$$

$$x = 16 \times 7$$

$$x = 112 \checkmark$$